

STUDY GUIDE

AWS IAM Policies

Assignment 19 — Writing managed and inline IAM policies, using condition keys, grouping permissions, and debugging a broken JSON policy.

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Assignment	19
Topic	AWS IAM Security
Format	Hands-on Lab

- AWS IAM
- S3
- EC2
- DynamoDB

Table of Contents

Task 1 — Create Read-Only S3 Policy	3
Step 1. Create S3 Bucket	3
Step 2. Create Policy	3
Step 3. Create User	4
Task 2 — Inline Policy (EC2 Dev Only)	5
Task 3 — Managed Policy + Group	6
Task 4 — Fix JSON Error	7
Task 5 — UserProfiles Policy	9

TASK 1

Create Read-Only S3 Policy

Step 1: Create S3 Bucket

AWS → S3 → Create bucket

Bucket configuration

Field	Value
Bucket name	my-music-playlists-673034950588
Region	ap-south-1

Click **Create bucket**.

Step 2: Create Policy

IAM → Policies → Create policy → JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ReadOnlyPlaylists",
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:ListBucket"
      ],
      "Resource": [
        "arn:aws:s3::my-music-playlists-673034950588",
        "arn:aws:s3::my-music-playlists-673034950588/*"
      ]
    }
  ]
}
```

Policy Name: `PlaylistReadOnly`

Step 3: Create User

IAM → Users → Create user

User configuration

Field	Value
User name	playlistViewer

Attach **PlaylistReadOnly** policy.

● RESULT

User can only read playlist objects.

TASK 2

Inline Policy (EC2 Dev Only)

Step 1: Create User

Create new IAM user: `devOpsIntern`

Step 2: Add Inline Policy

User → Permissions → Add permissions → Create inline policy → JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "StartStopDevEC2",
      "Effect": "Allow",
      "Action": [
        "ec2:StartInstances",
        "ec2:StopInstances"
      ],
      "Resource": "*",
      "Condition": {
        "StringEquals": {
          "ec2:ResourceTag/Environment": "Dev"
        }
      }
    }
  ]
}
```

Policy Name: `DevEC2Inline`

Meaning

What this policy allows

Action	Allowed
Start EC2	✓ Dev only
Stop EC2	✓ Dev only
Production	X Denied

TASK 3

Managed Policy + Group

Step 1: Create Group

IAM → User Groups → Create group

Group Name: `appReviewers`

Step 2: Create Managed Policy

IAM → Policies → Create Policy → JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ViewDynamoTables",
      "Effect": "Allow",
      "Action": [
        "dynamodb:ListTables",
        "dynamodb:DescribeTable",
        "dynamodb:Scan"
      ],
      "Resource": "*"
    }
  ]
}
```

Policy Name: `DynamoReviewAccess`

Attach it to `appReviewers` group.

Permissions

What each permission is used for

Permission	Use
<code>ListTables</code>	View tables
<code>DescribeTable</code>	View schema
<code>Scan</code>	Read records

TASK 4

Fix JSON Error

Given (Wrong)

BROKEN POLICY

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::user-uploads/*"
    }
  ]
}
```

PROBLEM

User cannot list bucket.

Correct JSON

FIXED POLICY

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "s3:ListBucket",
      "Resource": "arn:aws:s3:::user-uploads"
    },
    {
      "Effect": "Allow",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::user-uploads/*"
    }
  ]
}
```

● **FIX**

Added `s3:ListBucket` .

TASK 5

UserProfiles Policy

Create policy in IAM → JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "UpdateOwnProfile",
      "Effect": "Allow",
      "Action": [
        "dynamodb:GetItem",
        "dynamodb:UpdateItem"
      ],
      "Resource": "arn:aws:dynamodb:ap-south-1:673034950588:table/UserProfiles",
      "Condition": {
        "ForAllValues:StringEquals": {
          "dynamodb:LeadingKeys": [
            "${aws:username}"
          ]
        }
      }
    }
  ]
}
```

● TIP

The `dynamodb:LeadingKeys` condition restricts each user to only their own row — the partition key must match `${aws:username}`.